

Python Typecasting

(i) Other Datatype to Int

```
In [1]: 110.34
```

```
Out[1]: 110.34
```

```
In [2]: int(110.34)  # Float to int
```

```
Out[2]: 110
```

```
In [3]: int(True)    # Boolean to int
```

```
Out[3]: 1
```

```
In [4]: int(False)
```

```
Out[4]: 0
```

```
In [5]: int(10+20j)  # Complex to int
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[5], line 1  
----> 1 int(10+20j)  
  
TypeError: int() argument must be a string, a bytes-like object or a real number,  
not 'complex'
```

```
In [6]: int('10')    # No.string to int
```

```
Out[6]: 10
```

```
In [7]: int('Ten')   # Text string to int
```

```
-----  
ValueError                                Traceback (most recent call last)  
Cell In[7], line 1  
----> 1 int('Ten')  
  
ValueError: invalid literal for int() with base 10: 'Ten'
```

(ii) Other Datatype to Float

```
In [8]: float(110)    # Int to float
```

```
Out[8]: 110.0
```

```
In [9]: float(True)   # Boolean to float
```

Out[9]: 1.0

In [10]: `float(False)` *# Boolean to float*

Out[10]: 0.0

In [1]: `float(false)`

```
-----  
NameError                                Traceback (most recent call last)  
Cell In[1], line 1  
----> 1 float(false)  
  
NameError: name 'false' is not defined
```

In [12]: `float(10+20j)` *# Complex to float*

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[12], line 1  
----> 1 float(10+20j)  
  
TypeError: float() argument must be a string or a real number, not 'complex'
```

In [13]: `float('10')` *# No.string to float*

Out[13]: 10.0

In [14]: `float('Ten')` *# Text string to float*

```
-----  
ValueError                                Traceback (most recent call last)  
Cell In[14], line 1  
----> 1 float('Ten')  
  
ValueError: could not convert string to float: 'Ten'
```

(iii) Other Datatypes to Boolean

In [15]: `bool(2)` *# Int to boolean*

Out[15]: True

In [16]: `bool(0)`

Out[16]: False

In [24]: `bool(-2)`

Out[24]: True

In [17]: `bool(2.3)` *# Float to boolean*

Out[17]: True

In [18]: `bool(10+20j)` *# Complex to boolean*

Out[18]: True

```
In [19]: bool('10')      # No.string to boolean
```

Out[19]: True

```
In [20]: bool('Ten')     # Text string to boolean
```

Out[20]: True

```
In [21]: bool()          # Empty space
```

Out[21]: False

```
In [22]: bool( )
```

Out[22]: False

(iv) Other Datatypes to Complex

```
In [25]: complex(2)      # Int to complex
```

Out[25]: (2+0j)

```
In [26]: complex(3,4)
```

Out[26]: (3+4j)

```
In [27]: complex(3,4,5)
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[27], line 1  
----> 1 complex(3,4,5)  
TypeError: complex() takes at most 2 arguments (3 given)
```

```
In [28]: complex(2.3)    # Float to complex
```

Out[28]: (2.3+0j)

```
In [29]: complex(2.3,4.5)
```

Out[29]: (2.3+4.5j)

```
In [30]: complex(True)   # Boolean to complex
```

Out[30]: (1+0j)

```
In [31]: complex(False)
```

Out[31]: 0j

```
In [32]: complex('10')   # No.string to complex
```

Out[32]: (10+0j)

In [33]: `complex('Ten')` *# Text string to complex*

```
-----  
ValueError                                Traceback (most recent call last)  
Cell In[33], line 1  
----> 1 complex('Ten')  
  
ValueError: complex() arg is a malformed string
```

In [34]: `complex('10','20')`

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[34], line 1  
----> 1 complex('10','20')  
  
TypeError: complex() argument 'real' must be a real number, not str
```

In [35]: `complex(10, '20')`

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[35], line 1  
----> 1 complex(10, '20')  
  
TypeError: complex() argument 'imag' must be a real number, not str
```

In [36]: `complex('10',20)`

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[36], line 1  
----> 1 complex('10',20)  
  
TypeError: complex() argument 'real' must be a real number, not str
```

(v) Other Datatypes to String

In [37]: `str(10)` *# Int to string*

Out[37]: '10'

In [39]: `str(0)`

Out[39]: '0'

In [38]: `str(10.4)` *# Float to string*

Out[38]: '10.4'

In [40]: `str(True)` *# Boolean to string*

Out[40]: 'True'

In [41]: `str(False)`

Out[41]: 'False'

In [42]: `str(10+20j)` *# Complex to string*

Out[42]: '(10+20j)'

In []: